INTRODUCTION TO INFORMATION GEOMETRY

Population Proportions and Replicator Dynamics

Week 1 · Lecture 1

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Counts and proportions

Two observations of a population:

	Type A	Type B	Total
First census	48	72	120
Second census	90	180	270



Which type gained in number? Which gained in share?

Counts and proportions

Two observations of a population:

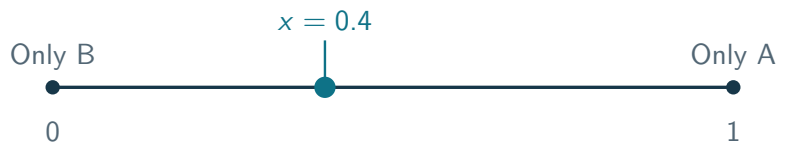
	Type A	Type B	Total
First census	48	72	120
Second census	90	180	270



Both types grew in number. Type B gained in share.

One variable describes the composition

$$x(t) = \frac{N_A(t)}{N_A(t) + N_B(t)}, \qquad 1 - x(t) = \frac{N_B(t)}{N_A(t) + N_B(t)}.$$



The state space of the proportion is $[0, 1]$.

Many populations share one composition

$$(N_A, N_B) = (20, 30), \qquad (200, 300), \qquad (2000, 3000)$$

$$x = \frac{N_A}{N_A + N_B} = \frac{2}{5}$$

Multiplying both counts by the same positive number leaves x unchanged.

The proportion describes relative abundance while the total $N = N_A + N_B$ records population size.

Growth equations for the two types

Let f_A and f_B be the per-capita growth rates.

$$\boxed{N_A' = f_A N_A \qquad N_B' = f_B N_B}$$

- $f_A > 0$ means type A increases in number.
- $f_A < 0$ means type A decreases in number.
- The units of each f_i are inverse time.

Differentiating the proportion

Write $N = N_A + N_B$, so $x = N_A / N$.

$$x' = \frac{N'_A N - N_A N'}{N^2}.$$

$N' = ?$ $\qquad$ $x' = ?$

Differentiating the proportion

Write $N = N_A + N_B$, so $x = N_A / N$.

$$x' = \frac{N'_A N - N_A N'}{N^2}.$$

$$N' = f_A N_A + f_B N_B$$

$$x' = \frac{N_A N_B}{N^2}(f_A - f_B).$$

The two-strategy replicator equation

$$x' = x(1 - x)(f_A - f_B)$$

x	$1 - x$	$f_A - f_B$
Current share of A	Current share of B	Relative fitness advantage

The sign of x' describes which type is gaining proportion.

Relative advantage controls direction

For an interior state, $0 < x < 1$, the factor $x(1-x)$ is positive.

$$f_A > f_B \implies x' > 0$$

$$f_A < f_B \implies x' < 0$$

$$f_A = f_B \implies x' = 0$$

The two counts may both increase or both decrease. Their *difference in per-capita rates* determines the change in composition.

A declining population

At a particular time, suppose

$$x = 0.4, \qquad f_A = -0.1, \qquad f_B = -0.4 \quad \text{(per day)}.$$

Find the instantaneous rate of change of the proportion:

$$x' = x(1-x)(f_A - f_B) = _____.$$

Then calculate the total population's per-capita growth rate

$$\frac{N'}{N} = xf_A + (1-x)f_B = _____.$$

A declining population

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Find the instantaneous rate of change of the proportion:

$$x' = x(1-x)(f_A - f_B) = 0.072 \text{ day}^{-1}.$$

Then calculate the total population's per-capita growth rate

$$\frac{N'}{N} = xf_A + (1-x)f_B = -0.28 \text{ day}^{-1}.$$

Type A gains share while the total population declines.

Constant fitness advantage

Suppose the difference $f_A - f_B = k$ is constant.

$$x' = kx(1 - x).$$

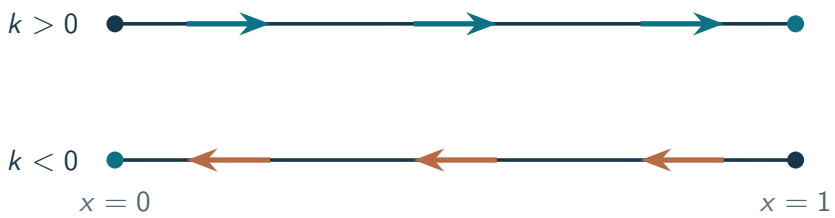
- $k > 0$: A increases its share from every interior state.
- $k < 0$: B increases its share from every interior state.
- $k = 0$: every composition is stationary.

Phase lines for constant advantage



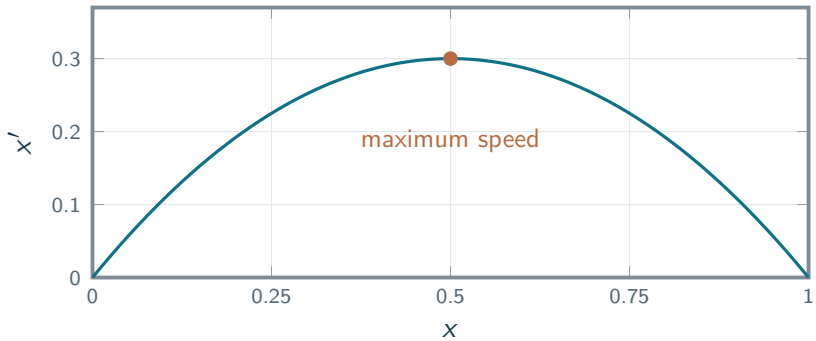
Draw the direction arrows. Which endpoint attracts for each sign of k?

Phase lines for constant advantage



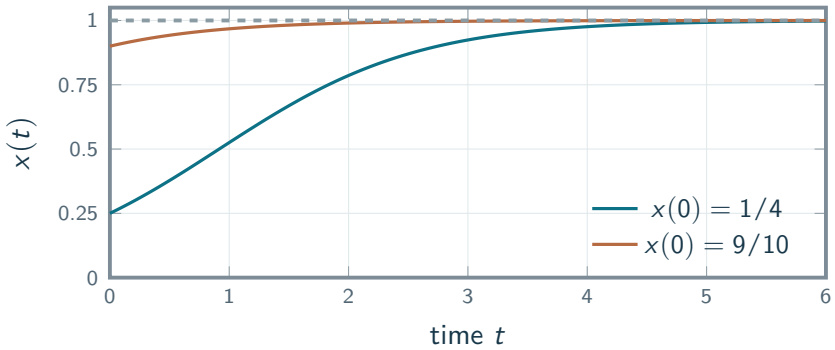
Filled teal endpoints attract trajectories from the interior.

The logistic velocity field



For $x' = 1.2x(1 - x)$, motion slows near either endpoint.

Solution curves in time



The phase line gives direction. The initial condition determines a particular curve.

Solving the logistic equation

For an interior solution of $x' = kx(1-x)$,

$$\frac{dx}{x(1-x)} = k\,dt.$$

Use partial fractions:

$$\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}.$$

$$\int \frac{dx}{x(1-x)} = \underline{\qquad\qquad\qquad\qquad}.$$

Solving the logistic equation

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$$\frac{dx}{x(1-x)} = k\,dt.$$

Use partial fractions:

$$\frac{1}{x(1-x)} = \frac{1}{x} + \frac{1}{1-x}.$$

$$\int \frac{dx}{x(1-x)} = \ln \frac{x}{1-x} = kt + C.$$

The explicit logistic solution

$$x(t) = \frac{1}{1 + \frac{1-x_0}{x_0} e^{-kt}}$$

For $0 < x_0 < 1$, $k > 0$ gives $x(t) \to 1$ as $t \to \infty$. The endpoint states $x_0 = 0$ and $x_0 = 1$ are constant solutions.

Time to an equal mixture

Suppose

$$x' = 1.2x(1-x), \qquad x(0) = \frac{1}{4}.$$

At what time does the proportion first reach $x = \frac{1}{2}$?

$$\frac{1}{2} = \frac{1}{1 + 3e^{-1.2t}}$$

$$t = _____.$$

Time to an equal mixture

Suppose

$$x' = 1.2x(1-x), \qquad x(0) = \frac{1}{4}.$$

At what time does the proportion first reach $x = \frac{1}{2}$?

$$\frac{1}{2} = \frac{1}{1 + 3e^{-1.2t}}$$

$$t = \frac{\ln 3}{1.2} \approx 0.916 \text{ time units.}$$

Composition and total growth

Two fitness pairs can give the same proportion equation:

$$(f_A, f_B) = (1, 0) \qquad \text{and} \qquad (2, 1).$$

Both have $f_A - f_B = 1$, hence both produce

$$x' = x(1 - x).$$

Their totals grow at different rates, because

$$\frac{N'}{N} = xf_A + (1 - x)f_B.$$

The composition equation retains relative growth information.